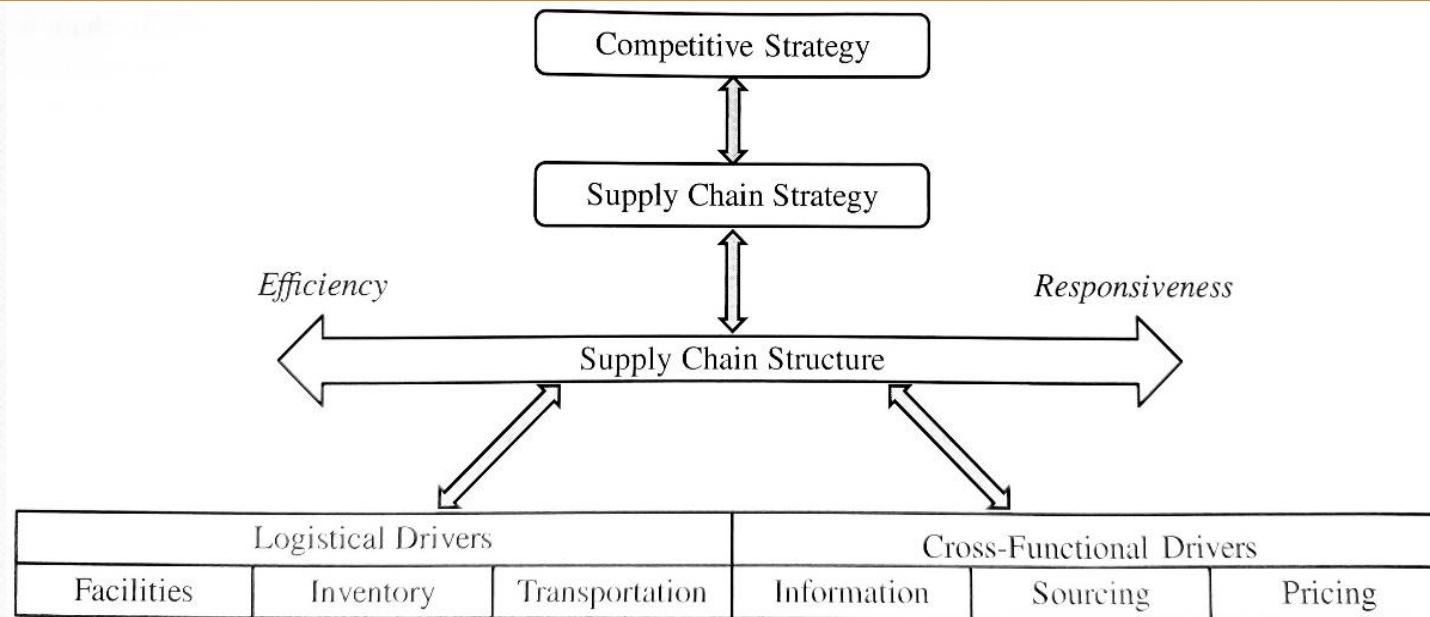


**POM 610-631:**  
Supply Chain Management I  
  
Walmart Business Case Presentation

PRESENTED BY: Dong, Yuhan; Ndong, Mathieu; Shelly, Henry

# Analysis of Walmart's Six Supply Chain Drivers

# The Relationship Between Supply Chain Operations and the Six Supply Chain Drivers



**Figure 3-1** Supply Chain Decision-Making Framework





# Facilities

(Logistical Driver)



- **Capability**
  - Supercenters function primarily as retail stores that house approximately 120,000 different products available for consumer purchase
  - Distribution Centers (DCs) receive and house approximately 77% of merchandise that ultimately ends up in supercenters
- **Location**
  - United States: supercenters are strategically located within 10 miles of 90% of the population
  - Using the “hub-and-spoke” design, DCs are strategically placed within an average of 130 miles from supercenters
- **Capacity**
  - Use of DCs, as well as plans for consolidation centers, allow supercenters to achieve lower capacity and inventory levels while still maintaining the responsive the supply chain strategy requires





# Inventory

(Logistical Driver)



- **Cycle Inventory**
  - Maintained low costs by purchasing merchandise in large batches, as a means of capitalizing on the economies of scale of its suppliers
- **Safe Inventory**
  - Avoided need for excess capacity through use of information systems and inventory data
- **Seasonal Inventory**
  - Again, avoided need for excess capacity through use of information systems and inventory data
- **Level of Product Availability**
  - Use of strategically placed DCs supported lower levels at retail supercenters



# Transportation

(Logistical Driver)



- **Design of Transportation Network**
  - Mix of “direct to demand point” shipping and “consolidation point” shipping
  - “Hub-and-spoke” distribution channel design promotes efficiency and responsiveness
- **Choice of Transportation Mode**
  - Walmart owned and operated truck fleet
  - Distribution sourced through common carriers





# Information

(Cross-Functional Driver)



- **Demand Planning**
  - Access to Retail Link allowed Walmart's suppliers to accurately estimate future demand forecasts based upon historical sales data
- **Coordination and Information Sharing**
  - The profitability of Walmart's total supply chain was maximized via the practice of information sharing
- **Sales and Operations Planning (S&OP)**
  - Walmart structure supply chain strategy to meet future demand forecasts through use of historical sales data





# Sourcing

(Cross-Functional Driver)



- **Combination of both “In-House” and “Outsourcing”**
  - In-House – (e.g.) distribution via Walmart’s privately owned and operated truck fleet
  - Outsourced – (e.g.) delivery of perishable grocery merchandise via common carriers
- **Supplier Selection**
  - Walmart estimated to have approximately 60,000 suppliers globally
  - Leverage advantage → Walmart
- **Procurement**
  - Walmart’s buyers worked with an array of suppliers
  - Buyers met quarterly to (e.g.) network, gather & exchange field intelligence, review new products, etc.



# Pricing

(Cross-Functional Driver)



- **Pricing and Economies of Scale**
  - Walmart promoted economies of scale for its suppliers, as well exploiting economies of scale for itself
- **Every Day Low Pricing (EDLP), rather than High-Low Pricing**
  - EDLP helped Walmart to keep its costs low while promoting a more stable demand for products
- **Menu Pricing, rather than Fixed Price**
  - Pricing contingent upon supplier's adherence to on-time delivery requirements
  - Price negotiations with suppliers

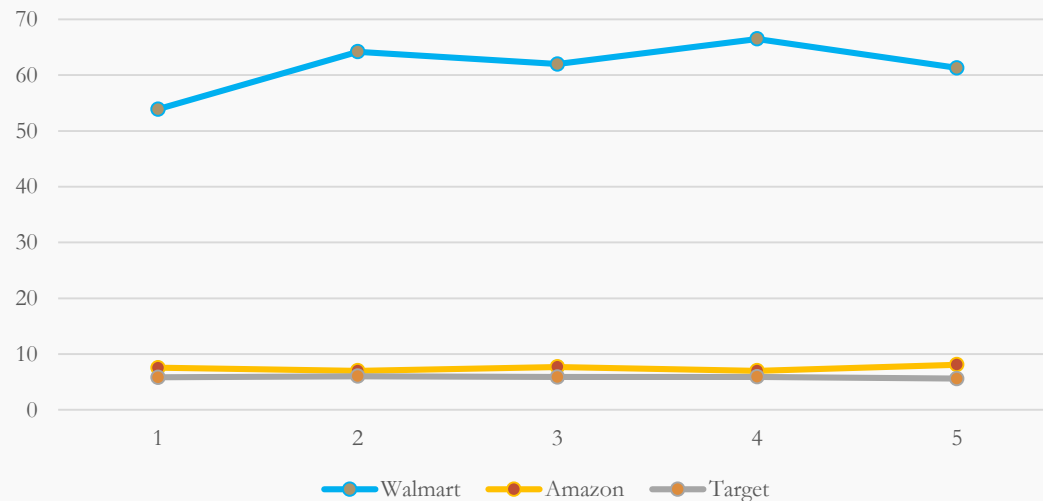


# Analysis of Walmart's Financial Performance

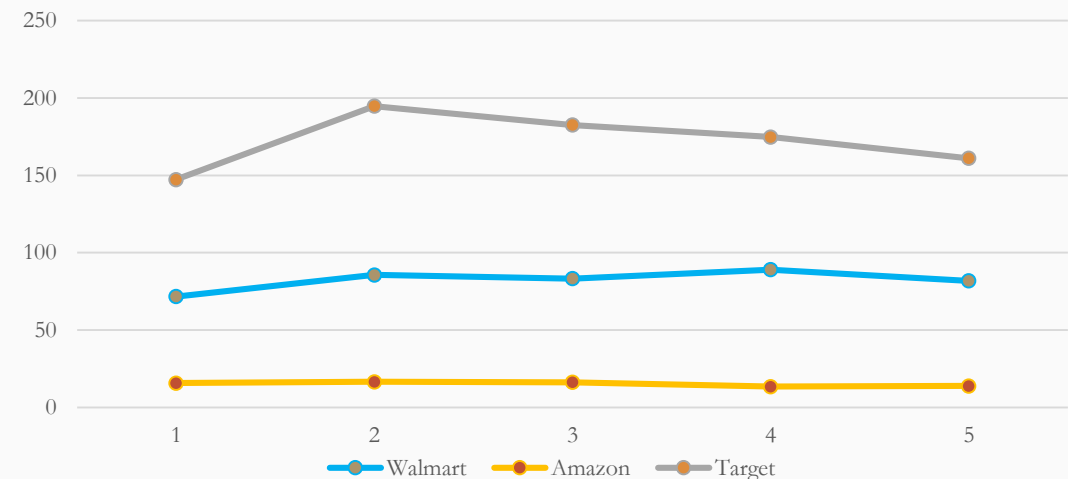
Walmart VS. Amazon VS. Target Study

# Supply Chain Metrics: INVT and ART

## Inventory Turnover (INVT)



## Accounts Receivable Turnover (ART)

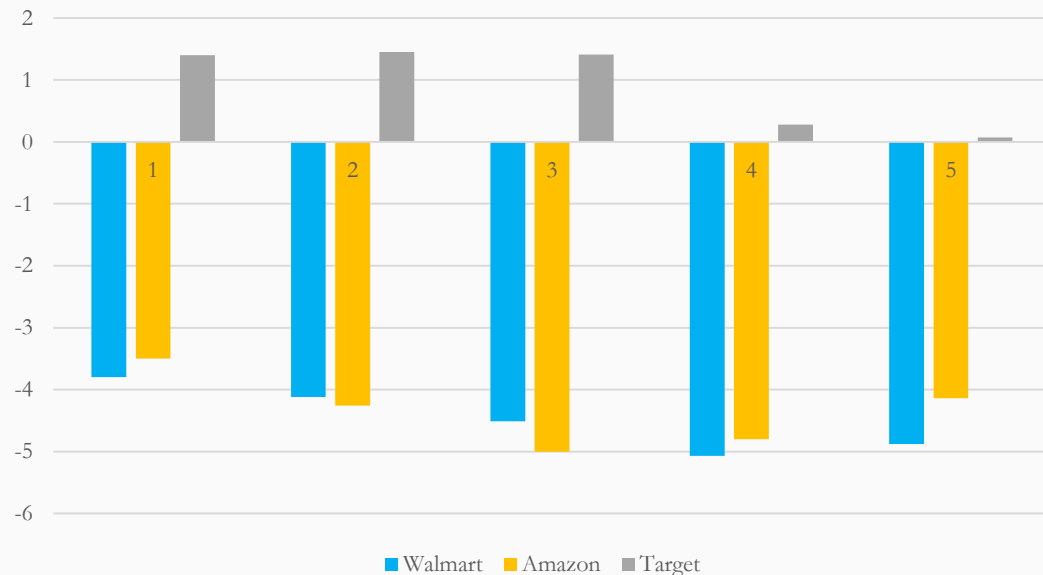


- Walmart turns its inventory ten times faster than Target on average
- Walmart ART values are good relative to competitors
- Both ART and INVT values have decreased from 2018 to 2019 for Walmart

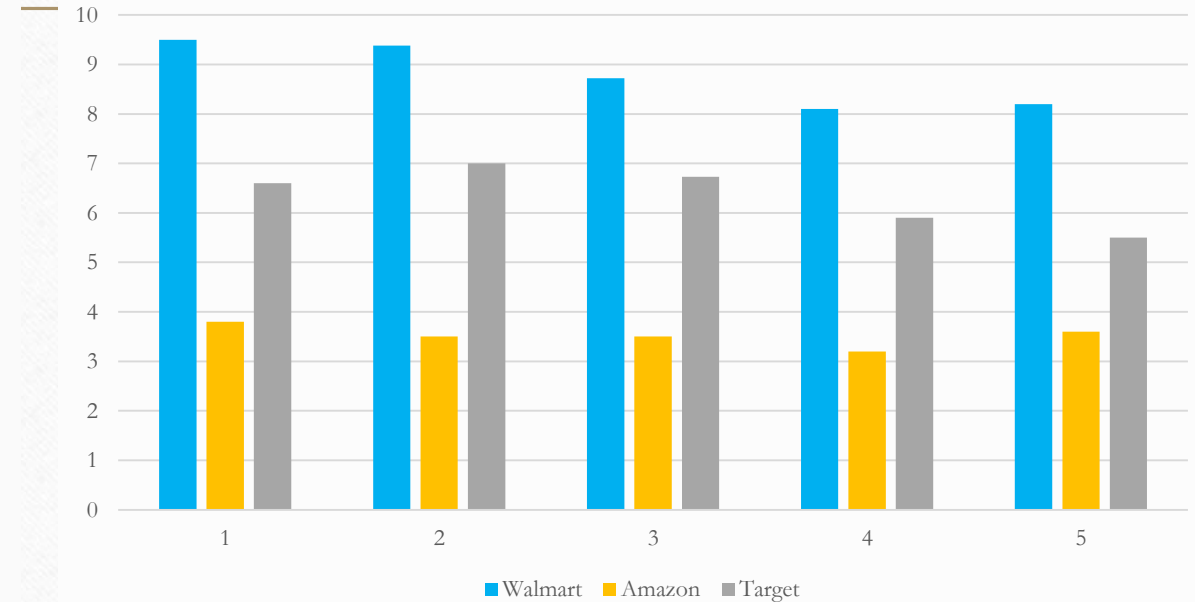


# Supply Chain Metrics: APT and C2C

Cash-to-Cash Cycle (C2C)



Account Payable Turnover (APT)



- Walmart and Amazon have almost the same value of -5 in 2018. It means that they collected their money from the sales of their products more than 5 weeks before they had to pay their respective suppliers
- Walmart finance its own operations for 6.4 weeks in 2019 with its suppliers' money compare to 9.45 weeks for Amazon.

# Operating Margin and Net Margin

Net Margin Table

| Year | Walmart      | Amazon | Target |
|------|--------------|--------|--------|
| 2015 | <b>0.03</b>  | -0.003 | 0.03   |
| 2016 | <b>0.03</b>  | 0.005  | 0.05   |
| 2017 | <b>0.028</b> | 0.02   | 0.04   |
| 2018 | <b>0.02</b>  | 0.02   | 0.04   |
| 2019 | <b>0.01</b>  | 0.04   | 0.038  |

Operating Margin Table

| Year | Walmart     | Amazon | Target |
|------|-------------|--------|--------|
| 2015 | <b>0.06</b> | 0.02   | 0.06   |
| 2016 | <b>0.05</b> | 0.039  | 0.07   |
| 2017 | <b>0.05</b> | 0.03   | 0.07   |
| 2018 | <b>0.04</b> | 0.07   | 0.06   |
| 2019 | <b>0.04</b> | 0.06   | 0.05   |

- Net margin shows how good a company is at converting revenue into profits
- Steady decrease from 2015 to 2019
- The Operating margin for Amazon is increasing over this period indicates a company whose profitability is improving.
- Steady decrease from 2015 to 2019



# Gross Profit Return, Asset Turnover & ROE

**Gross profit Return on Inventory**

**Asset Turnover**

**ROE**

| Year | Walmart     | Amazon | Target | Year | Walmart     | Amazon | Target | Year | Walmart     | Amazon | Target |
|------|-------------|--------|--------|------|-------------|--------|--------|------|-------------|--------|--------|
| 2015 | <b>2.67</b> | 3.16   | 2.43   | 2015 | <b>2.38</b> | 1.63   | 1.75   | 2015 | <b>0.19</b> | -0.02  | 0.17   |
| 2016 | <b>2.72</b> | 3.45   | 2.53   | 2016 | <b>2.41</b> | 1.63   | 1.83   | 2016 | <b>0.19</b> | 0.04   | 0.25   |
| 2017 | <b>2.89</b> | 4.16   | 2.57   | 2017 | <b>2.44</b> | 1.63   | 1.88   | 2017 | <b>0.17</b> | 0.12   | 0.24   |
| 2018 | <b>2.89</b> | 4.1    | 2.49   | 2018 | <b>2.45</b> | 1.35   | 1.86   | 2018 | <b>0.12</b> | 0.11   | 0.25   |
| 2019 | <b>2.91</b> | 5.45   | 2.32   | 2019 | <b>2.35</b> | 1.43   | 1.82   | 2019 | <b>0.09</b> | 0.23   | 0.26   |

- High asset Turnover
- Steady increase in Gross Profit Return on Inventory
- Decrease in ROE
- Walmart is less efficient in generating profit

# Analysis of Walmart's Supply Chain Distribution Structures and Challenges Moving Forward



# Distribution Options

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- 1. Manufacturer storage, direct shipping
- 2. Manufacturer storage, direct shipping, in-transit merge
- 3. Distributor/retailer storage, carrier delivery
- 4. Distributor/retailer storage, last-mile delivery
- 5. Manufacturer/distributor storage, customer pickup
- 6. Retail storage with customer pickup

# Manufacturer Storage with Direct Shipping

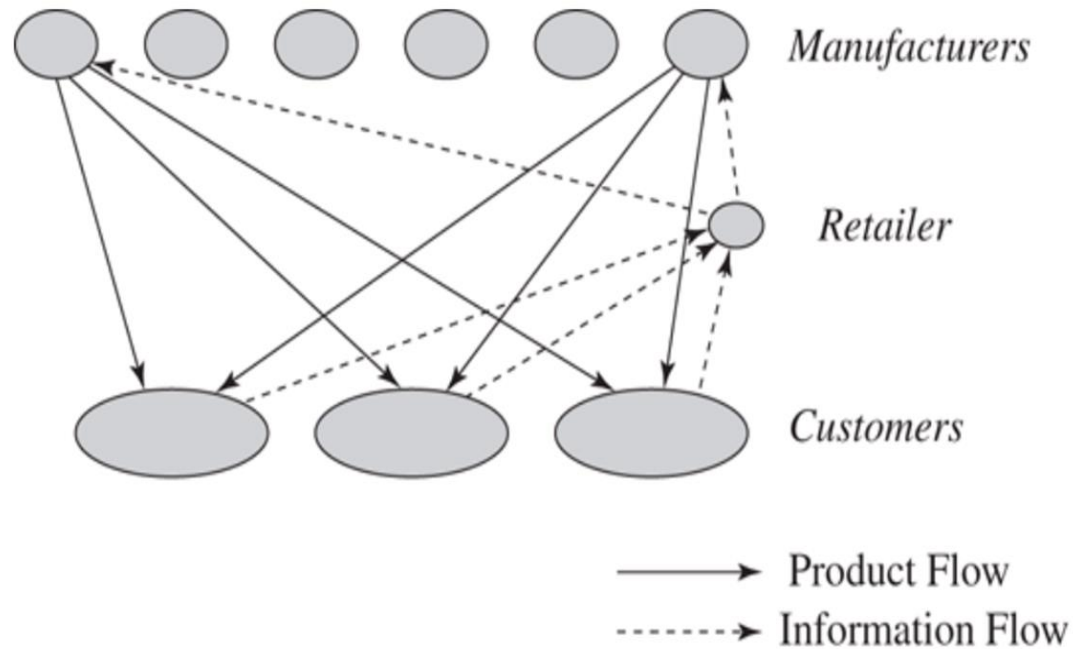
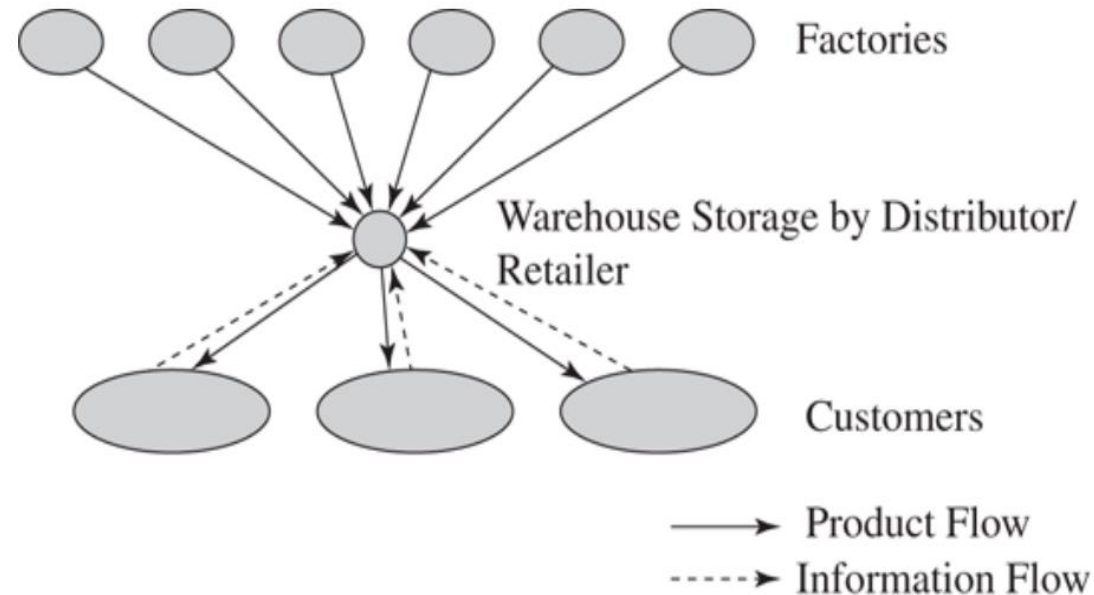


Figure 4-6 Manufacturer Storage with Direct Shipping



# Distributor Storage with Carrier Delivery



**Figure 4-8 Distributor Storage with Carrier Delivery**

# Manufacturer or Distributor Storage with Customer Pickup

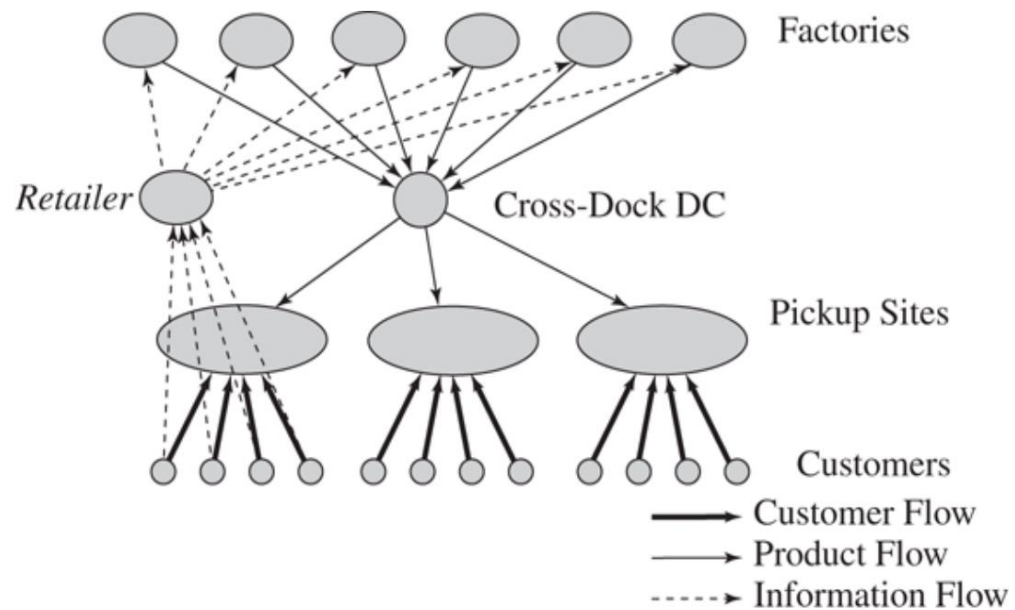


Figure 4-10 Manufacturer or Distributor Warehouse Storage with Consumer Pickup



# Challenges Moving Forward

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- What changes needed to make to Walmart's ever-expanding supply chain to support its strategic objectives.
- What supply chain capabilities would they need as its business model continued to evolve.

# Strategies Moving Forward

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- Reshaped its supply chain strategy to tackle the “last miles”
- Focus on opportunities to reduce operational and supply chain costs, which translated into lower retail prices.
- Insisted on saturating the area within a day’s driving distance of DC in order to gain economies of scale.
- Actively developed new apps for customers and store associates to convenient customer’s shopping experience and inventory management in stores.

ANY  
QUESTIONS  
?